IBNU THYRA MAGANI

Student Developer & Computational Systems Builder

📍 Semarang, Central Java, Indonesia ✉ ibnuthyra488@gmail.com 💻 github.com/LaSserafim
🌐 ibnu-thyra-magani-portfolio.vercel.app

Institution: BINUS SCHOOL Semarang (Cambridge Assessment International Education) | **Current Standing:** Grade 11 (Expected Graduation: June 2027) | **Student UID:** 2470005430 • **NISN:** 0086859561

EXECUTIVE PROFILE & ENGINEERING PHILOSOPHY

Honest Systems Architecture

> **Transparent AI-Assisted Engineering Workflow:** Rather than claiming multi-year manual low-level syntax mastery, my software engineering pairs rigorous theoretical problem formulation with modern AI leverage. I formulate the governing mathematical and physical equations (stochastic differential equations, kinematic biomechanics, extreme-value realized volatility estimators) and use LLMs (Claude, DeepSeek) for rapid boilerplate scaffolding, API bindings, and formula translation. I then personally debug runtime edge cases, resolve numerical convergence failures (e.g., Cox-Ross-Rubinstein continuity fixes), eliminate external dependencies (Abramowitz-Stegun rational approximations to remove SciPy runtime bloat), enforce zero-cloud client privacy via WebAssembly, and author automated verification test suites (15/15 passing tests).

STANDARDIZED TESTING & ENGLISH PROFICIENCY

Official College Board & IELTS Credentials

Digital SAT (College Board — Sep 12, 2026) — 1260 Composite (81st %ile)

Mathematics (90th Percentile)	690 / 800
• Algebra & Advanced Math	680–800 Band
• Problem-Solving, Data Analysis & Geometry	610–670 Band
Reading and Writing (65th Percentile)	570 / 800
• Information & Ideas	610–670 Band

IELTS Academic (TRF: 26ID507816MAGI017A) — Band 7.0 (CEFR C1)

Listening	7.0	Reading	7.5
Speaking	6.5	Writing	6.0

✓ Officially certified at CEFR C1 (Proficient User). Meets/exceeds direct admissions requirements for English-track STEM degree programs globally.

OFFICIAL ACADEMIC TRANSCRIPTS (BINUS SCHOOL SEMARANG)

Grade 10 & 11 Certified True Copies • Combined Cumulative Average: 85.49%

Grade 10 Official Report (AY 2024–2025) — Core Avg: 88.20% (Straight A/A+)

SUBJECT	SEM 1	SEM 2	FINAL	GRADE
Bahasa Indonesia	92.6	93.9	93.2	A+
Physics	**89.1**	**91.3**	**90.2**	**A**
Mathematics	**86.8**	**91.6**	**89.2**	**A**
Sociology	89.9	89.2	89.5	A
Chemistry	**91.8**	**86.1**	**88.9**	**A**
Geography	90.5	85.4	88.0	A
Business Studies	87.2	87.9	87.6	A
Biology	**90.6**	**80.8**	**85.7**	**A**
English Language	81.2	81.5	81.4	A

Grade 11 Progress Summary (AY 2025–2026) — Overall: 83.27% (STEM: 86.6%)

SUBJECT	SEM 1	SEM 2	FINAL	TRAJECTORY
Mathematics	**85.0**	**92.0**	**88.5**	+7.0 Upward
Physics	**88.0**	**89.0**	**88.5**	+1.0 Solid
Chemistry	**85.0**	**86.0**	**85.5**	+1.0 Solid
Biology	**82.0**	**86.0**	**84.0**	+4.0 Solid
PPKN (Civics)	89.0	93.0	**91.0**	+4.0 Gain
Bahasa Indonesia	85.0	85.0	**85.0**	Stable
Bahasa Jawa	82.0	89.0	**85.5**	+7.0 Gain
English Language	84.0	84.0	**84.0**	Bilingual
Physical Education	70.0	80.0	**75.0**	+10.0 Gain
Information Tech.	81.0	67.0	**74.0**	Practical dev
Islam Studies	78.0	74.0	**76.0**	Completed

* *Key Academic Takeaway:* Clear upward trajectory in Cambridge Advanced Mathematics (+7.0 points) and sustained high distinction across physical sciences (Grade 11 STEM average: 86.6%). Certified with Principal wet signature.

CORE TECHNICAL SKILLS & ENGINEERING STACK

Production Tooling & Systems

Programming Languages & Web Architecture: Python 3.12, JavaScript (ES6+ / ES Modules), TypeScript, HTML5, CSS3, FastAPI (ASGI backend), React 19, Vite, Tailwind CSS.

Scientific & Numerical Computing: NumPy (vectorized multi-dimensional matrix operations), SciPy, Pandas, Plotly.js, Chart.js, HTML5 Canvas API.

Mathematical & Quantitative Competencies: Geometric Brownian Motion (GBM), Itô's Lemma, Cox-Ross-Rubinstein (CRR) binomial lattice tree option pricing, Non-parametric Value-at-Risk (VaR), Conditional Value-at-Risk (CVaR / Expected Shortfall), Cornish-Fisher expansion for skewness/kurtosis adjustment, Parkinson (1980) High-Low & Garman-Klass (1980) realized volatility estimators, Corwin-Schultz (2012) bid-ask spread proxy, Kapandji-Hansraj cervical biomechanical load curves.

DevOps, Privacy & Infrastructure: Git/GitHub, GitHub Actions (CI/CD workflows), macOS launchd system daemons, Headless Chrome / Puppeteer (vector PDF compilation), MediaPipe Pose WebAssembly (client-side edge ML), Supabase, Vercel.

TECHNICAL INTERESTS & RESEARCH DOMAINS

Academic & Research Pillars

- **Probability & Stochastic Modeling:** Formulating continuous and discrete stochastic processes to quantify tail-risk bounds and simulate forward asset trajectories under market uncertainty.
- **Financial Econometrics & Market Microstructure:** Implementing empirical volatility and liquidity proxies from raw order book and OHLCV data without relying on synthetic or idealized assumptions.
- **Automated Data Pipelines:** Designing resilient, self-healing data extractors (ETL) with automated network recovery daemons, cache-based idempotency, and automated vector document pipelines.
- **Client-Side Privacy & Edge Inference:** Deploying computer vision and biometric evaluation pipelines entirely in-browser using WebAssembly, ensuring zero user media transmits across network boundaries.
- **Interactive Computational Simulations:** Translating formal academic syllabi (biochemistry, cellular kinetics) into deterministic state machines with 60 FPS procedural SVG animations.

FEATURED SOFTWARE ENGINEERING PROJECTS (1 / 2)

Quantitative & Financial Systems

FutureAs — Quantitative Equity Valuation & Stochastic Engine

`futureas.vercel.app` | `GitHub Repo`

Institutional-grade quantitative equity valuation, discrete option lattice trees, and stochastic modeling terminal.

`Python 3.12 • FastAPI • NumPy • SciPy • Yahoo Finance API • React • Tailwind CSS • Plotly`

- **Stochastic Engine:** Computes 10,000-path Geometric Brownian Motion (GBM) Monte Carlo forward trajectories across 252 trading days in under 80ms via vectorized NumPy cumulative log-return sums.
- **Discrete Option Lattice:** Programmed a Cox-Ross-Rubinstein (CRR) binomial lattice tree valuation model; personally derived and implemented the continuity correction to eliminate multi-step boundary convergence errors.
- **Downside Risk Analytics:** Formulated non-parametric 95% Value-at-Risk (VaR) and 95% Expected Shortfall (CVaR).
- **Dependency Elimination:** Implemented the Abramowitz & Stegun rational polynomial approximation for inverse normal CDF, removing external SciPy runtime overhead and accelerating serverless cold-start execution.
- **Full-Stack Architecture:** Engineered 7 analytical modules (Valuation, Trend, Lump-Sum vs DCA, Probability Cones, Drawdowns, SEC Form 4/144 Insiders, Crash Radar) verified with a 15/15 passing automated test suite (`test_api.py`, `test_quant.py`).

QUANT.IDX Terminal — Indonesian Equity Econometric Terminal

`quantidx.vercel.app` | `GitHub Repo`

Quantitative equity research and empirical market microstructure terminal for the Indonesia Stock Exchange (IDX / BEI).

`Python 3.12 • FastAPI • NumPy • Pandas • TradingView Pro Widget • Chart.js • Tailwind CSS`

- **Extreme-Value Volatility:** Implemented 3 realized volatility estimators from econometric literature: Parkinson (1980) High-Low (~5x statistical efficiency over close-to-close metrics), Garman-Klass (1980) OHLC, and Rogers-Satchell (1991).
- **Non-Gaussian Tail Risk:** Modeled downside risk via the Cornish-Fisher expansion, adjusting VaR and CVaR for empirical skewness and excess kurtosis; implemented closed-form Chi-Square p-values for Jarque-Bera (1987) normality tests.
- **Liquidity Proxies:** Modeled empirical market liquidity using the Corwin-Schultz (2012) bid-ask spread proxy from 2-day high/low price ratios and the Amihud (2002) Illiquidity Ratio.
- **Full-Stack Interface:** Built a high-density financial interface embedding live TradingView Pro charts and automated DeepSeek econometric memorandums with deterministic offline fallbacks.

FEATURED SOFTWARE ENGINEERING PROJECTS (2 / 2)

Computer Vision, Automation & Computational Biology

MediaSpine — In-Browser Clinical Posture Screener & Biometric Scanner

mediaspine.vercel.app | GitHub Repo

Real-time webcam posture screening tool built for SpineSafe NGO running entirely client-side via WebAssembly.

`Vanilla JavaScript (ES Modules) • HTML5 Canvas • MediaPipe Pose Wasm • DeepSeek V3 • Supabase • Vercel`

- **Zero-Cloud Privacy:** Runs MediaPipe BlazePose Lite locally via WebAssembly; frames are evaluated in memory and discarded immediately, guaranteeing 100% visual privacy with no cloud video uploads.
- **Automated Viewpoint Classifier:** Automatically separates Sagittal (profile) and Coronal (frontal) viewpoints based on dynamic horizontal bi-acromial (shoulder) landmark Euclidean distance thresholds.
- **Kinematics & Load Modeling:** Calculates 5 metrics (neck deviation angle, shoulder tilt, head tilt, 30-frame rolling stability variance, and 3-second hold stability gate). Modeled the Kapandji-Hansraj cervical spinal load curve (11 to 60 lbs).
- **Human Performance Engineering:** Personally implemented frame-drop throttling under GPU contention, circular buffers for landmark smoothing, and calibrated a deterministic 0–100 penalty scoring rubric.

Stock Intelligence Cloud — Autonomous Research & Dossier Pipeline

GitHub Repo

Automated daily equity research pipeline ingesting SEC filings and market data to compile and email PDF dossiers.

`Python 3.12 • Puppeteer (Headless Chrome) • GitHub Actions • macOS launchd • SMTP`

- **Dual-Layer Redundancy:** Orchestrated an automated execution pipeline running locally at 09:00 WIB via macOS `launchd` daemons, paired with an automated GitHub Actions cloud cron workflow as autonomous fallback.
- **Idempotency & Resilience:** Enforced delivery idempotency via GitHub cache keys to prevent duplicate emails; built an automated network verification daemon with exponential backoff retry loops to prevent launchd crashes upon waking from sleep.
- **Vector PDF Publishing:** Automated vector PDF compilation using Headless Chrome for Testing with precision CSS print media pagination.

The Link Reaction — Interactive Cambridge 9700 Biological Simulation

the-link-reaction.vercel.app | GitHub Repo

Interactive 8-stage cellular respiration simulation aligned with Cambridge International A-Level Biology (9700 Section 12).

`HTML5 • CSS3 Keyframe Engine • Vanilla JavaScript (ES Modules) • Procedural SVG`

- **Biochemical State Engine:** Programmed an 8-stage finite state transition engine enforcing exact biological reaction steps: mitochondrial transport, decarboxylation (CO_2 release), dehydrogenation ($NAD^+ \rightarrow NADH$), and Coenzyme A (CoA-SH) sulfhydryl linkage.
- **Stoichiometric Conservation:** Enforced stoichiometric carbon conservation tracking 3C pyruvate splitting into 2C acetyl and 1C CO_2.
- **Zero-Dependency Performance:** Replaced heavy 3D WebGL libraries with procedural SVG transforms and CSS keyframes, ensuring zero external dependencies and 60 FPS performance across all low-power classroom devices.

HONORS, COMPETITIONS & EXTRACURRICULAR LEADERSHIP

Verified Extracurricular Record

- **Silver Medal — Physics Engineering:** BISF (Bandung International Science Fair). Awarded for applied physics research, quantitative reasoning, and experimental mechanical engineering analysis.
- **Cambridge Academic Honors:** Straight A / A+ across all 9 examinable core subjects in Grade 10 (88.20% composite academic average).
- **Academic Resilience in Advanced Mathematics:** Demonstrated an upward performance trajectory in Grade 11 Cambridge Mathematics, advancing from 85.0 (Semester 1) to 92.0 (Semester 2).
- **School Band Elective:** Graded **A+**; active instrumentalist in school ensemble and public performances.
- **Athletics & Community Discipline:** Active member of BINUS School Badminton Club; maintained 95–100% attendance throughout high school career.